

Xutao Mao

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[Website](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

- **City University of Hong Kong** Hong Kong
Ph.D. Student, Computer Science Sep 2026 – May 2030 (Expected)
 - Research Areas: Agent Safety, Trustworthy LLM/VLM, Mechanistic Interpretability
- **Vanderbilt University** Nashville, US
Bachelor of Science, Honors in Computer Science & Mathematics Aug 2022 – Dec 2025
 - Research Areas: LLM, Data Mining

RESEARCH EXPERIENCE

- **City University of Hong Kong** Hong Kong
Graduate Researcher | Advisor: Dr. Cong Wang Aug 2026 – Present
 - Acting and coordinating: how AI fails while they act in the world [AHA] [CAVE-Bench] [TrustFork]
 - Remembering and improving: how AI failures persist [PASB] [MisEvolve]
 - Looking inside and stepping in: how AI failures happens & how to mitigate failures [STARE] [Memory Interpret]
- **Hong Kong Baptist University** Hong Kong
Research Assistant | Advisor: Dr. Bo Han Dec 2025 – Aug 2026
 - Focus: CoT Monitorability [TAME]
- **Vanderbilt University** Nashville, US
Undergraduate Researcher | Advisor: Dr. Tyler Derr Feb 2025 – Dec 2025
 - Focus: LLM; Data Mining [Review Recommendation]

PUBLICATIONS

C=CONFERENCE, P=PREPRINT

- [P.1] Xutao Mao, Rui Qian, Xiang Zheng[†], Cong Wang[‡]. **Agent Hacks Agents: Autoresearch Discovers Vulnerabilities in Production Agents.**
- [P.2] Xutao Mao*, Rui Qian*, Longxiang Wang, Xinjian Yi, Mingxuan Li, Linghan Chen, Yudong Gao, Xiang Zheng[†], Cong Wang[‡]. **You're Right, Let Me Fix It: How LLM Agents Damage Correct Work When Falsely Accused.**
- [P.3] Xutao Mao*, Rui Qian*, Linghan Chen, Yudong Gao, Junchi Liao, Jiulin Cai, Jinman Zhao, Cong Wang[‡]. **Trust the Brand, Lose Control: How Identity Hijacks LLM Agent Orchestration.**
- [P.4] Xutao Mao, Liangjie Zhao, Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han[‡], Xiang Zheng[†], Cong Wang[‡]. **Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Self-Improving Personal Agents.**
- [P.5] Xutao Mao, Liangjie Zhao, Xiang Zheng[†], Cong Wang[‡]. **Practice Makes Unsafe: Skill Misevolution in Self-Improving LLM Agents.**
- [P.6] Leyao Wang*, Xutao Mao*, Xuhui Zhan, Yuying Zhao, Bo Ni, Ryan A. Rossi, Nesreen K. Ahmed, Tyler Derr[‡]. **Towards Bridging Review Sparsity in Recommendation with Textual Edge Graph Representation.**
- [P.7] Xutao Mao, Jinman Zhao, Gerald Penn, Cong Wang[‡]. **What Happens Inside Agent Memory? Circuit Analysis from Emergence to Diagnosis.**
- [C.1] Xutao Mao*, Jianing Zhu*, Jinman Zhao, Tongliang Liu, Xiaowen Chu, Cong Wang[‡], Bo Han[‡]. **Taming CoT Obfuscation in VLMs: From Mechanistic Evidence to Activation-Level Enforcement.** *NeurIPS 2026 (Poster).*
- [C.2] Haobo Zhang*, Xutao Mao*, Guangyuan Dong, Ziwei Li[‡], Xuanbo Su, Kaijie Chen, Jing Yang, Zheng Lin. **MemMark: State-Evolution Attribution Watermarking for Agent Long-Term Memory Systems.** *EMNLP 2026 (Finding).*
- [C.3] Xutao Mao, Liangjie Zhao, Tao Liu, Xiang Zheng[†], Hongying Zan, Cong Wang[‡]. **STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack.** *ICML 2026 (Poster).*
- [C.4] Xutao Mao[‡], Ezra Tao, Leyao Wang. **MindVote: When AI Meets the Wild West of Social Media Opinion.** *AAAI 2026 (Oral).*
- [C.5] Tao Liu[†], Xutao Mao[†], Hongying Zan[‡], Dixuan Zhang, Yifan Li, Haixin Liu, Lulu Kong, Jiaming Hou, Rui Li, YunLong Li, Aoze Zheng, Zhiqiang Zhang, Zhewei Luo, Kunli Zhang, Min Peng. **LogicCat: Text-to-SQL Benchmark for Multi-Domain Reasoning Challenges.** *AAAI 2026 (Poster).*

Notes: *equal contribution; [†] alphabetical co-first/co-equal as noted; [‡]corresponding author.

WORK EXPERIENCE

- **Pegasystems**
Software Development Engineer Intern
- Cambridge, US*
May 2024 – Aug 2024

HONORS AND AWARDS

- **Hong Kong Postgraduate Scholarship**
City University of Hong Kong
- 2026
- **Vanderbilt Summer Research Program Scholarship**
Vanderbilt University
- 2025

PROFESSIONAL SERVICE

- **Conference Reviewer:** ICLR '27; NeurIPS '26; AAAI '26 / '27; TheWebConf '26

TECHNICAL SKILLS & INTERESTS

- **Models & Training:** LLM/VLM architectures (MoE, RoPE, FlashAttention); SFT, RL post-training (PPO, GRPO, DPO, RLHF), agentic RL, LoRA; PyTorch, DeepSpeed, FSDP, vLLM, verl; multi-node multi-GPU training
- **Research Interests:** Agent Safety, Interpretability.